Mathematical Models in Biotechnology

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Abstract

Predicting the intracellular flux distribution of a growing cell is a central problem in biotechnology, yet the internal fluxes, kinetic parameters, and metabolite concentrations that would determine it are, at genome scale, largely unmeasured, while the stoichiometry of the reaction network is fully known. This chapter develops flux balance analysis, the constraint-based method that turns stoichiometry and conservation into a linear program for the flux distribution, as an integrative framework in which the flux bounds, rather than the stoichiometric matrix, carry the biological information that narrows a prediction. The flux constraint is derived from an open mole balance on a growing culture rather than asserted as a steady state, a derivation that retains the growth-dilution term omitted from the textbook form and makes explicit the operating regimes, fed-batch and cell-free among them, where that term cannot be neglected. Each flux bound is then written as a product of dimensionless factors that integrate thermodynamic reversibility, enzyme kinetics, gene expression, and regulation in a common mathematical object. Supplying a thermodynamic estimate, a turnover number, an expression measurement, or a regulatory model therefore refines the corresponding factor in the bound. Two worked steady-state examples, a urea-cycle metabolic network constrained entirely from public databases and a feedback-inhibited pathway whose regulated throughput is recovered by incorporating regulatory information against an independent kinetic reference model, make the construction concrete, and two published systems, a dynamic cell-free reaction and a model-guided glycan-producing strain, show these information layers combined at genome scale. Estimating a flux distribution from data and designing one by intervention emerge as the same operation on the bounds, run in opposite directions.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a target product, bioprocess development tunes feeding and environment to sustain it, and strain design asks which genetic changes will move a phenotype in a wanted direction; each is, at bottom, a question about the intracellular flux distribution, the rates at which the reactions of metabolism actually run under a given condition [1]. That distribution is what a quantitative model is asked to predict, and it is also difficult to observe. A genome sequence and the stoichiometry of its annotated reactions are knowable in broad outline, though real reconstructions carry their own uncertainty in reaction membership, directionality, compartmentalization, and biomass composition; the internal fluxes, the kinetic parameters that would set them, and the metabolite concentrations that drive them are, at genome scale, far less constrained still. The problem is therefore to predict a metabolic phenotype from what can be known rather than from what a direct calculation would require.

Two broad strategies answer that problem. Mechanistic kinetic models write an explicit rate law for every reaction, from Michaelis–Menten saturation [2] through mass action to the power-law formalism of biochemical systems theory [3, 4], and integrate the resulting equations forward in time; they resolve the dynamics in full but demand a rate constant, an affinity, or a kinetic order for every step, parameters that are rarely available beyond a handful of well-studied pathways. Constraint-based models make the opposite trade. They ask only for the stoichiometry of the network and impose conservation of mass at steady state, and in doing so they scale to genome-wide reconstructions of thousands of reactions [1, 5]. Flux balance analysis is the constraint-based workhorse: it casts phenotype prediction as a linear program over the flux vector [6, 7]. Conservation alone, however, does not determine a flux distribution; it fixes a whole subspace of them, and what selects a prediction from that subspace is the box of upper and lower bounds placed on each reaction. Flux bounds are a major interface through which condition-specific thermodynamic, kinetic, expression, and regulatory information enters a constraint-based model, and the question this chapter takes up is how it enters: through which bound does a thermodynamic estimate, a turnover number, an expression measurement, or a regulatory signal actually change a flux prediction?

This chapter develops flux balance analysis as an integrative framework, organized around bounds that provide a natural mechanism for combining different kinds and levels of biological information in a flux estimate. The flux constraint itself is derived not by asserting steady state but from an open mole balance on a growing culture, a derivation that keeps the growth-dilution term omitted from the textbook form and writes the constraint as $\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}$ rather than $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$. A single bound is then read as a product of factors representing thermodynamic reversibility, enzyme kinetics, gene expression, and regulation. Refining a factor means supplying the data or mechanistic model that determines it. Two worked steady-state examples make the construction concrete: a urea-cycle metabolic network whose flux distribution is pinned by thermodynamic and kinetic data drawn entirely from public databases, and a feedback-inhibited pathway whose regulated throughput, generated independently from a kinetic reference model, is recovered by incorporating the allosteric activity factor in a single bound. Two reported systems then show the same information layers combined at a scale no hand calculation could reach, one estimating fluxes in a dynamic cell-free reaction [8] and one designing a glycan-producing strain by model-guided gene deletion [9]. Read across these cases, estimating a flux distribution from data and designing one by intervention evaluate the bounds the same way once an intervention is chosen; design differs in also requiring a combinatorial search, over candidate deletions or expression changes, for which bounds to move [10].

2 From Mole Balances to the Flux Constraint

Flux balance analysis (FBA) uses an algebraic constraint derived from a mole balance. Writing the balance out identifies the assumptions used to obtain its familiar textbook form and the operating conditions under which that reduction applies. Consider a well-mixed bioreactor of volume V holding a growing cell culture. The culture exchanges mass with its surroundings through a set of physical streams s (feed, harvest, gas sparging) and hosts an intracellular reaction network with n fluxes $\hat{v}_j$, $j = 1, \ldots, n$, acting on m metabolite species. For a species i present at concentration C_i, an open mole balance over the whole culture accounts for material entering and leaving through the streams together with material produced or consumed by the reaction network:

$$\sum_s d_s\, C_{i,s}\, \dot{V}_s \;+\; \sum_j \sigma_{ij} \hat{v}_j\, V \;=\; \frac{d}{dt}\big(C_i V\big), \tag{1}$$

where $d_s = +1$ for streams entering the reactor and $d_s = -1$ for streams leaving it, $C_{i,s}$ is the concentration of species i in stream s, $\dot{V}_s$ is the volumetric flow rate of that stream, and σ_{ij} is the stoichiometric coefficient of species i in reaction j, the (i, j) entry of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m\times n}$. This mole balance, Equation (1), does not require assumptions about growth rate, feeding strategy, or steady state and provides the starting point for the reduction below.

Two steps turn Equation (1) into a statement about fluxes alone. First, write the reactor volume in specific units, $V = B\bar{V}$, with B the total biomass in the reactor and $\bar{V}$ the culture volume per unit biomass; this ties the extensive balance to the biomass that is actually growing, and the specific growth rate follows as $\mu \equiv (1/B)\, dB/dt$. Second, the product rule expands the accumulation term as:

$$\frac{d}{dt}\big(C_i V\big) = \frac{d}{dt}\big(C_i B\bar{V}\big) = B\bar{V}\,\frac{dC_i}{dt} + C_i\bar{V}\,\frac{dB}{dt} + C_i B\,\frac{d\bar{V}}{dt}. \tag{2}$$

Dividing Equation (1) through by $V = B\bar{V}$ and substituting Equation (2) isolates the intensive balance for species i:

$$\frac{1}{B\bar{V}}\sum_s d_s\, C_{i,s}\,\dot{V}_s \;+\; \sum_j \sigma_{ij}\hat{v}_j \;=\; \frac{dC_i}{dt} \;+\; \mu\, C_i \;+\; \frac{C_i}{\bar{V}}\,\frac{d\bar{V}}{dt}, \tag{3}$$

where the middle term on the right, μC_i with $\mu = (1/B)\, dB/dt$, is the growth-dilution term produced by the product rule. Three assumptions each remove one of the remaining terms. If the intracellular pool of species i is at pseudo-steady state ($dC_i/dt = 0$, the usual separation-of-timescales argument, since metabolite turnover on the order of seconds to minutes is fast relative to a doubling time of hours), if the specific culture volume is constant ($d\bar{V}/dt = 0$, so the working volume tracks the growing biomass rather than drifting relative to it), and if species i crosses no physical stream (the sum over s vanishes for this species), the intensive balance reduces to the growth-dilution constraint:

$$C_i\,\mu \;=\; \sum_j \sigma_{ij}\hat{v}_j \qquad\Longleftrightarrow\qquad \boxed{\mathbf{S}\hat{\mathbf{v}} \;=\; \mu\mathbf{x}}, \tag{4}$$

where $\mathbf{x} = (C_1, \ldots, C_m)^\top$ collects the intracellular concentrations and $\hat{\mathbf{v}} \in \mathbb{R}^n$ collects the fluxes. The resulting flux constraint, Equation (4), retains growth dilution: net metabolic production of each intracellular species equals its dilution rate in the growing culture.

The form used throughout the constraint-based modeling literature is obtained by neglecting the right-hand side of Equation (4):

$$\mathbf{S}\hat{\mathbf{v}} \;=\; \mathbf{0}. \tag{5}$$

The steady-state relation in Equation (5) is a special case of Equation (4), rather than an independent starting point [6]. It is an appropriate approximation when the two conditions that justify neglecting $\mu\mathbf{x}$ hold together: growth is slow relative to metabolic turnover, and intracellular pools are dilute enough that $\mu\mathbf{x}$ stays small in absolute terms even when μ itself is not negligible. Both conditions can fail in relevant operating regimes. In fed-batch culture the feed is metered to increase the working volume over the course of a run, so the specific volume drifts ($d\bar{V}/dt \neq 0$); the assumption used to reach Equation (4) no longer holds, and the omitted term carries information about the feed schedule that Equation (5) cannot see. In cell-free systems there is no growing cell to dilute into ($\mu = 0$), yet the reaction mixture remains time varying: transcript, protein, and metabolite pools accumulate and decay over the course of a batch reaction ($d\mathbf{x}/dt \neq 0$), so the pseudo-steady-state assumption does not apply. A cell-free capstone later in this chapter returns to this setting, where the accumulation term omitted by Equation (5) is measured directly.

The dilution-retaining constraint and its steady-state special case, Equations (4) and (5), are, as written, statements about a closed system: every species that appears has its production and consumption accounted for entirely by the internal network $\mathbf{S}\hat{\mathbf{v}}$, with the physical stream terms of Equation (1) already omitted along the way. Cellular metabolism is open, since cells take up substrates and secrete products across the boundary of whatever system is being modeled. Constraint-based reconstructions restore that openness without reintroducing the stream terms explicitly: hypothetical exchange, or boundary, reactions are added to the network itself, one pseudo-reaction $\emptyset \rightleftharpoons M_i$ for each boundary metabolite M_i, entering $\mathbf{S}$ as an isolated nonzero column with no reaction partner [1]. Flux through an exchange reaction enters Equation (4) or Equation (5) exactly as any metabolic flux does, but it is now interpretable as uptake or secretion across the system boundary rather than as a step of intracellular chemistry. A closed algebraic statement can thereby represent a genuinely open system: openness becomes entirely a matter of which reactions are admitted into $\mathbf{S}$ and how their bounds are set, the question taken up next.

3 The Linear Program and Its Geometry

The flux constraint retaining growth dilution, Equation (4), together with its steady-state special case Equation (5), is a statement of conservation: it does not by itself pick out any particular flux distribution among all those consistent with it. Turning conservation into a prediction requires two further ingredients. The first is a box of flux bounds, $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$, one lower and one upper bound per reaction, encoding thermodynamic reversibility ($\ell_j < 0$ permitted), enzyme capacity, and substrate availability; how those bounds are set is taken up next. The second is an objective, a linear functional $\mathbf{c}^\top \hat{\mathbf{v}}$ of the flux vector that states what the cell, or the modeler, is presumed to optimize. With both in hand the mole-balance constraint becomes a linear program:

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top \hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \quad (\text{or } \mu\mathbf{x}), \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{6}$$

This linear program is the complete textbook formulation of flux balance analysis [6]. The objective vector $\mathbf{c}$ is a choice, not a derived quantity: setting $c_j = 1$ at the index of a biomass pseudo-reaction, one that drains precursors in the ratios required to build new cell mass, casts the LP as growth-rate maximization, the default assumption for a cell under selection pressure to divide. Setting $c_j = 1$ instead at the index of a product-export exchange reaction casts the same LP as yield maximization for a chosen metabolite, the framing relevant to a bioprocess engineer optimizing a production strain rather than modeling wild-type physiology. These cases have the same constraint set and differ only in $\mathbf{c}$.

Linear programming addresses the underdetermination of the constraint set. A genome-scale reconstruction lists every reaction annotated to a genome, and the reaction count n typically exceeds the metabolite count m, because a single metabolite typically participates in several reactions while a single reaction typically touches only a handful of metabolites. The stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ therefore has many more columns than rows, and its null space $\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}$ is nontrivial: a whole subspace, not a point, satisfies the steady-state constraint. Every flux vector in $\mathcal{N}(\mathbf{S})$ conserves mass at every internal node exactly as well as every other; conservation alone cannot distinguish among them. The flux bounds cut this unbounded subspace down to a bounded polytope, and the objective in Equation (6) selects a single vertex, or a face, of that polytope, but the underlying indeterminacy, that most of the space allowed by stoichiometry is not resolved by stoichiometry, remains after an optimum has been found. This progression from conservation to bounds to optimization is summarized in Figure 1.

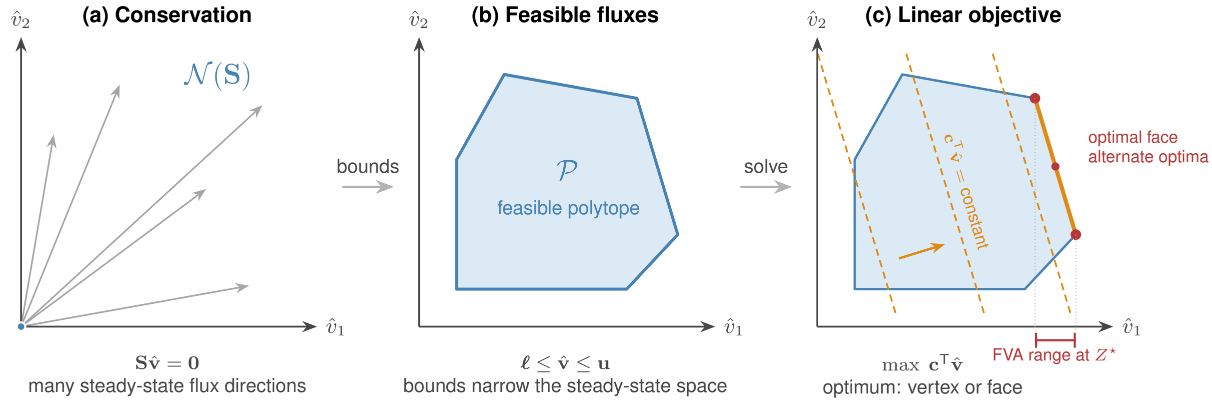

Figure 1: Geometry of flux balance analysis in a two-flux projection. (a) Conservation restricts steady-state fluxes to the null space $\mathcal{N}(\mathbf{S})$; the rays represent admissible steady-state directions in the displayed projection. (b) Lower and upper flux bounds intersect that space to form the convex feasible polytope $\mathcal{P}$. (c) Parallel level sets of the linear objective $\mathbf{c}^\top\hat{\mathbf{v}}$ move in the objective direction until they support the polytope at an optimum. That optimum may be a unique vertex or an entire face. In the latter case, fluxes can vary while retaining the same objective value, and flux variability analysis projects the optimal face onto a reaction-specific interval, illustrated here for $\hat{v}_1$.

The singular value decomposition makes the geometry of that indeterminacy explicit. Writing $\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, with $\mathbf{U} \in \mathbb{R}^{m\times m}$ and $\mathbf{V} \in \mathbb{R}^{n\times n}$ orthogonal and $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ diagonal with nonnegative entries, the columns of $\mathbf{V}$ associated with zero singular values span $\mathcal{N}(\mathbf{S})$, the right null space: every feasible steady-state flux mode is a linear combination of these columns, and reading Equation (6) through this lens, the LP is a search over coefficients of that combination rather than over the raw flux vector. The columns of $\mathbf{U}$ associated with zero singular values span the left null space of $\mathbf{S}$, equivalently the null space of $\mathbf{S}^\top$; a vector $\mathbf{y}$ in this space defines a conserved moiety, a linear combination $\mathbf{y}^\top\mathbf{x}$ of metabolite concentrations, such as a bound cofactor pair sharing a fixed pool total, that is invariant under the reaction network regardless of which flux vector in $\mathcal{N}(\mathbf{S})$ is realized. For a genome-scale reconstruction the rank of $\mathbf{S}$ is smaller than the reaction count, $\operatorname{rank}(\mathbf{S}) \ll n$, so the right null space, and with it the space of flux distributions the constraints alone cannot distinguish, is large [1]. That indeterminacy can be read reaction by reaction. Because an LP optimum need not be unique, and because many reactions can vary freely without changing the objective value at all, flux variability analysis reports a range rather than a point for each flux: with the objective fixed at its optimal value $Z^\star$ (or at a specified fraction of it), each v_j is in turn minimized and maximized subject to Equation (6)'s constraints [7]. A reaction whose minimum and maximum coincide is pinned by the network topology and bounds at the optimal objective value used for that calculation, while a reaction with a wide range is left essentially free; a different objective can produce a different feasible face and a different range for the same reaction, and the pattern of which fluxes fall into which category is exactly the information that a single optimal flux vector, taken alone, cannot report.

4 Integrative Flux Bounds

Conservation alone leaves the flux distribution underdetermined: the mole-balance constraint fixes a subspace, not a point, and the box of bounds together with the objective is what narrows that subspace to a prediction. The objective is a modeling choice. The bounds are where biology enters, and they enter not as numbers to be looked up but as a model in their own right. Writing a single upper bound as a product of factors makes that model visible. For reaction j, the factorized bound is:

$$-\delta_j \Big[V^{\circ}_{\mathrm{max},j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot) \Big] \;\leq\; \hat{v}_j \;\leq\; V^{\circ}_{\mathrm{max},j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot), \tag{7}$$

where $V^{\circ}_{\mathrm{max},j}$ sets a reference capacity carrying the units of flux and the remaining factors are dimensionless corrections that scale that capacity up or down. Each factor represents a distinct kind and level of biological information: δ_j encodes thermodynamic directionality, $V^{\circ}_{\mathrm{max},j}$ and f_j encode enzyme kinetics, (e/e°) encodes the abundance established by gene expression, and θ_j encodes allosteric regulation of the activity of the enzyme molecules already present. The expression and translation controls u_j and w_j therefore do not replace θ_j in this bound: they act through the transcript and protein balances below and ultimately change (e/e°). A bound therefore provides an explicit accounting of what is known about a reaction. With no information beyond a capacity scale the correction factors are set to unity, and each is refined as measurements or mechanistic models become available. This is also where the underdetermined polytope is finally narrowed: not by conservation but by the bounds, and every informative factor tightens that polytope using evidence of a particular kind, so that tracing a prediction back to its cause is a matter of asking which factor in Equation (7) was responsible.

The thermodynamic factor is the switch δ_j, which decides whether the reaction may carry flux in the reverse direction at all. It takes one of two values:

$$\delta_j \;=\; \begin{cases} 1, & \text{reaction } j \text{ reversible},\\ 0, & \text{reaction } j \text{ irreversible}, \end{cases} \tag{8}$$

so that $\delta_j = 1$ leaves the lower bound of Equation (7) negative and the reaction free to run backward, while $\delta_j = 0$ collapses the lower bound to zero and pins the reaction to the forward direction. The assignment follows the sign of $\Delta G^{\circ}_j - \Delta G^{*}_j$, the standard free energy of reaction measured against a threshold ΔG^{*}_j meant to represent the physiological concentration range, or equivalently a cutoff on the equilibrium constant $K_{\mathrm{eq},j}$: a reaction whose driving force cannot plausibly change sign anywhere in that range is declared irreversible, and its reverse flux is forbidden. This is a heuristic directionality rule, a stand-in for propagating a stated concentration range through $\Delta G = \Delta G^{\circ} + RT \ln Q$ and checking directly whether the sign can flip, not a substitute for that calculation where the range is actually known. Standard free energies for this test are drawn from group- and component-contribution estimates, for which eQuilibrator provides genome-scale coverage [11].

When the reaction is allowed to proceed, its capacity is set by enzyme kinetics, and this is the role of the reference scale $V^{\circ}_{\mathrm{max},j}$ together with the saturation factor f_j. The scale is a maximal velocity, given by the product of a turnover number and a characteristic enzyme abundance:

$$V^{\circ}_{\mathrm{max},j} \;=\; k^{\circ}_{\mathrm{cat},j}\, e^{\circ}, \tag{9}$$

where $k^{\circ}_{\mathrm{cat},j}$ is the catalytic rate constant of the enzyme catalyzing reaction j and e° is a reference enzyme concentration. Turnover numbers are tabulated per enzyme and substrate in BRENDA [12], and a representative e° can be read from compilations of cellular abundances such as BioNumbers

[13]. The scale $V^{\circ}_{\max,j}$ is the flux the enzyme would sustain at reference abundance and saturating substrate, and the fraction of that ceiling actually reached at the prevailing substrate concentration is the saturation factor. That factor follows from the enzyme mechanism itself. A single-substrate enzyme binds substrate reversibly and turns it over to product according to the mechanism:

$$E + S_j \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} [ES] \xrightarrow{k_{\text{cat}}} E + P, \tag{10}$$

with $[ES]$ the enzyme-substrate complex. Holding the complex at quasi-steady state, on the standard argument that it equilibrates fast relative to the depletion of substrate, and eliminating the free enzyme through the conservation of total enzyme $[E]_T = [E] + [ES]$, gives the complex as a hyperbolic function of substrate alone, $[ES] = [E]_T\,[S_j]/(K_{M,j} + [S_j])$ with $K_{M,j} \equiv (k_{-1} + k_{\text{cat}})/k_1$. The reaction velocity is the rate of the turnover step, $v_j = k_{\text{cat}}[ES]$, so writing the capacity as $V_{\max,j} = k_{\text{cat}}[E]_T$ recovers the Michaelis–Menten rate law [2], and the saturation factor is obtained by normalizing the rate to that capacity:

$$v_j \;=\; V_{\max,j}\,\frac{[S_j]}{K_{M,j} + [S_j]} \qquad \Longrightarrow \qquad f_j([S_j]) \;=\; \frac{v_j}{V_{\max,j}} \;=\; \frac{[S_j]}{K_{M,j} + [S_j]} \;\in [0,1], \tag{11}$$

with $[S_j]$ the substrate concentration and $K_{M,j}$ the Michaelis constant. The saturation factor f_j interpolates between negligible capacity ($f_j \to 0$ as substrate is exhausted) and the maximal capacity ($f_j \to 1$ at saturation), and it is the same hyperbolic law that governs single-enzyme assays. Because f_j depends on the substrate concentration, incorporating this factor makes the bound a function of the metabolic state rather than a fixed number, the point at which a static flux balance begins to acquire the state-dependence that a dynamic treatment carries to its conclusion.

The scale $V^{\circ}_{\max,j}$ was fixed at a reference enzyme abundance e°, but the abundance actually present is itself a variable set by gene expression, and the ratio (e/e°) is the factor through which that variability enters. To treat abundance as a modeled quantity rather than a fixed parameter, write balances for the transcript m_j and the protein p_j of the gene encoding the enzyme:

$$\dot{m}_j = r_{X,j}\,u_j - (\theta_{m,j} + \mu)\,m_j + \lambda_j, \tag{12}$$

$$\dot{p}_j = r_{L,j}\,w_j\,m_j - (\theta_{p,j} + \mu)\,p_j, \tag{13}$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation rate capacities, u_j and w_j are dimensionless control functions that modulate those capacities, and $\theta_{m,j}$ and $\theta_{p,j}$ are first-order degradation rate constants for message and protein, and λ_j is a basal, or leak, source of transcript. The loss term in each balance is not the degradation constant alone but the sum $(\theta_{m,j} + \mu)$ or $(\theta_{p,j} + \mu)$: every intracellular species is removed both by its own turnover and by dilution into the growing culture at the specific growth rate μ. That μ is not a new parameter. It is the very growth-dilution term retained in the constraint retaining growth dilution, Equation (4), carried through consistently: an enzyme is diluted by growth exactly as a metabolite is, and the same μ that balances metabolic production in the flux constraint sets the residence time of the transcripts and proteins that fix the flux bounds. Holding the balances at steady state gives the closed-form abundances:

$$m^{\star}_j = \frac{r_{X,j}u_j + \lambda_j}{\theta_{m,j} + \mu}, \qquad p^{\star}_j = \frac{r_{L,j}w_j\,m^{\star}_j}{\theta_{p,j} + \mu}, \qquad (e/e^{\circ}) = \frac{p^{\star}_j}{e^{\circ}}, \tag{14}$$

and the protein level $p^{\star}_j$ is mapped onto the abundance of the catalytic activity for reaction j through the gene-protein-reaction (GPR) rules of the reconstruction, which encode whether an activity requires a single gene product, several assembled into a complex, or any one of several interchangeable isozymes. Through Equation (14) the expression factor couples the flux ceiling to the entire upstream apparatus of transcription and translation, with μ standing in every denominator as the visible signature of the growth-aware bookkeeping begun in the mole-balance derivation. The ratio (e/e°) is also the natural port for measured data: a transcriptomic or proteomic profile enters here, rescaling the capacity of each reaction by the abundance actually observed under a given condition and turning one reconstruction into a family of condition-specific models.

One statistical mechanical construction can represent allosteric activity, transcription, and translation. The common idea is to classify configurations by function. Some configurations of an enzyme, promoter, or translation-initiation complex permit the process of interest; the remaining configurations do not. The physical configurations and the criterion for activity differ among the three processes, but the calculation is the same: the numerator sums the configurations that permit the process, while the denominator sums all configurations available to the regulated element.

Let $\mathcal{S}_q$ denote the set of configurations of a regulated element associated with process q, where q may be catalysis, transcription, or translation. Configuration $s \in \mathcal{S}_q$ has energy ϵ_s, with its statistical weight given by:

$$\omega_s(\mathbf{x}) = g_s(\mathbf{x})e^{-\beta\epsilon_s}, \qquad Z_q(\mathbf{x}) = \sum_{r\in\mathcal{S}_q} \omega_r(\mathbf{x}), \qquad p_s(\mathbf{x}) = \frac{\omega_s(\mathbf{x})}{Z_q(\mathbf{x})}, \tag{15}$$

where $\beta = 1/(k_BT)$ and Z_q is the partition function. The factor $g_s(\mathbf{x})$ accounts for ligand concentrations, binding stoichiometry, and degeneracy. For example, a configuration containing n bound molecules of an effector x contributes a concentration factor proportional to $(x/K)^n$. The partition function is therefore a concentration-dependent sum over every configuration, not only those that are functional.

Now let $\mathcal{A}_q \subseteq \mathcal{S}_q$ be the configurations that permit process q, and let $\mathcal{I}_q = \mathcal{S}_q \setminus \mathcal{A}_q$ be the configurations that do not. The dimensionless control for that process is the probability of occupying the active subset:

$$\Phi_q(\mathbf{x}) = \sum_{s\in\mathcal{A}_q} p_s(\mathbf{x}) = \frac{\displaystyle\sum_{s\in\mathcal{A}_q} g_s(\mathbf{x})e^{-\beta\epsilon_s}}{\displaystyle\sum_{r\in\mathcal{S}_q} g_r(\mathbf{x})e^{-\beta\epsilon_r}} = \frac{Z_{\mathcal{A}_q}}{Z_{\mathcal{A}_q} + Z_{\mathcal{I}_q}}. \tag{16}$$

Here $Z_{\mathcal{B}} = \sum_{s\in\mathcal{B}} \omega_s$ denotes a partition sum restricted to the subset $\mathcal{B}$. This ratio makes the functional classification explicit. Every active configuration appears in both the numerator and denominator; inactive configurations appear only in the denominator. Consequently $0 \le \Phi_q \le 1$, and an effector increases or decreases the control according to whether it preferentially weights active or inactive configurations.

The compact constants used in promoter models can be connected directly to these Boltzmann weights. Choose configuration 0 as a reference and write $\Delta\epsilon_s = \epsilon_s - \epsilon_0$. The relative energetic weight and the complete state weight are then related by:

$$K_{\star,s} \equiv \frac{\omega_s/g_s}{\omega_0/g_0} = e^{-\beta\Delta\epsilon_s}, \qquad \frac{\omega_s}{\omega_0} = \frac{g_s(\mathbf{x})}{g_0(\mathbf{x})} K_{\star,s}. \tag{17}$$

With the conventional choice $\omega_0 = g_0 = 1$, this becomes $\omega_s = g_s(\mathbf{x})K_{\star,s}$. Thus, a term such as $K_2 f_{TL}$ combines an energetic factor $K_2 = e^{-\beta\Delta\epsilon_2}$ with the availability f_{TL} of the ligand-bound transcription factor. In a fitted promoter model, however, K_1 and K_2 are effective, dimensionless weights: they may absorb a constant RNA-polymerase concentration, standard-state factors, and other lumped contributions, and need not equal measured dissociation constants.

Figure 2 summarizes the construction from molecular configurations to process-specific control:

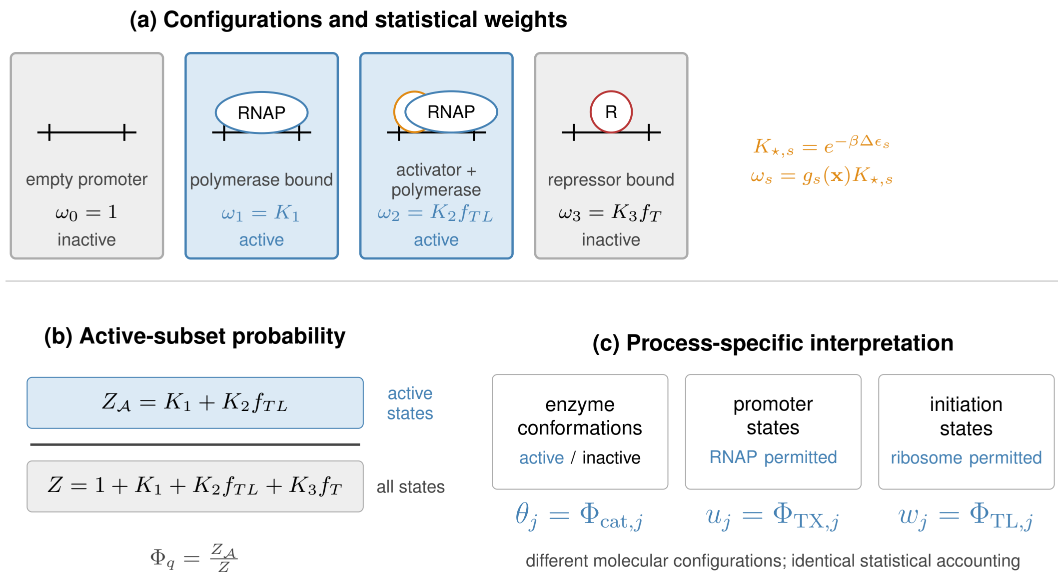

Figure 2: Active-subset construction of biological control factors. (a) Configuration weights combine relative energetics with regulator availability; blue states permit the process and gray states do not. (b) Active states enter the numerator, while all states enter the partition-function denominator. The promoter shown is schematic. (c) Specifying the relevant configurations gives allosteric activity θ_j, transcriptional control u_j, or translational control w_j from the same statistical accounting.

The inducible-promoter models of Moon, Voigt, and coworkers provide a concrete example of this construction [14]. Their Supplementary Figure S7 and Equations 3–6 enumerate promoter configurations and assign a partition-function term to each one. For the Lux promoter, the configurations have weights 1, K_1, and $K_2 f_{TL}$, corresponding respectively to an empty promoter, RNA-polymerase binding, and joint binding of RNA polymerase with the ligand-bound LuxR complex. The two RNA-polymerase-bound configurations permit transcription and therefore appear in the numerator. For the Tac promoter, the empty, RNA-polymerase-bound, and repressor-bound configurations have weights 1, K_1, and $K_2 f_T$; only the RNA-polymerase-bound configuration permits transcription. After normalization by the maximal promoter activity, the two controls are therefore given by:

$$u_{\mathrm{Lux}} = \frac{K_1 + K_2 f_{TL}}{1 + K_1 + K_2 f_{TL}}, \qquad u_{\mathrm{Tac}} = \frac{K_1}{1 + K_1 + K_2 f_T}. \tag{18}$$

Here f_{TL} is the ligand-bound fraction of the transcription factor, whereas f_T is its ligand-free fraction. The Lux example adds weight to an active configuration as inducer increases; the Tac example removes weight from an inactive, repressor-bound configuration. Both are direct specializations of Equation (16).

A two-configuration element illustrates the reduction. If an inactive configuration has weight $\omega_I = 1$ and an effector-stabilized active configuration has weight $\omega_A = (x/K)^n e^{-\beta\Delta\epsilon}$, then the control becomes:

$$\Phi_q(x) = \frac{(x/K)^n e^{-\beta\Delta\epsilon}}{1 + (x/K)^n e^{-\beta\Delta\epsilon}}, \tag{19}$$

where $\Delta\epsilon$ is the active-state energy relative to the inactive reference. The familiar Hill-like form is thus a normalized active-state weight. If the effector instead stabilizes an inactive configuration, its concentration-dependent weight enters the denominator but not the numerator, and the control decreases with x.

The process-specific factors used above are instances of Equation (16):

$$\theta_j = \Phi_{\text{cat},j}, \qquad \bar{u}_j = \Phi_{\text{TX},j}, \qquad w_j = \Phi_{\text{TL},j}. \tag{20}$$

The allosteric factor θ_j multiplies catalytic capacity directly in Equation (7). The total transcriptional control $\bar{u}_j$ is separated below into regulated and basal contributions, while w_j multiplies translational capacity in Equation (13). These dimensionless controls are distinct from the degradation-rate constants $\theta_{m,j}$ and $\theta_{p,j}$.

Basal expression follows by subdividing the transcriptionally active set. Let $\mathcal{A}_{\text{TX}}^{\text{reg}}$ contain active configurations reached through the regulated input and let $\mathcal{A}_{\text{TX}}^{\text{basal}}$ contain configurations that support transcription without that input. Their contributions are:

$$\bar{u}_j = \frac{Z_{\mathcal{A}_{\text{TX}}^{\text{reg}}}}{Z_{\text{TX}}} + \frac{Z_{\mathcal{A}_{\text{TX}}^{\text{basal}}}}{Z_{\text{TX}}} = u_j + u_j^{\dagger}, \qquad \lambda_j \equiv r_{X,j} u_j^{\dagger}. \tag{21}$$

The regulated contribution u_j enters the transcription term in Equation (12), whereas the basal contribution $u_j^{\dagger}$ supplies its leak source λ_j. The promoter-state specialization follows the construction used by Moon, Voigt, and coworkers [14]. The effective biophysical model of Adhikari and coworkers extends the same active-subset accounting to transcriptional and translational control [15]; the factorization here places those controls alongside the corresponding allosteric factor.

Each information layer requires data that is often unavailable: thermodynamic estimates, turnover numbers, substrate concentrations, expression measurements, and regulatory parameters. When that data is scarce the correction factors are set to unity, $f_j \approx 1$, $(e/e^{\circ}) \approx 1$, and $\theta_j \approx 1$, and the general bound collapses to the two-parameter box:

$$-\delta_j \, V_{\text{max},j}^{\circ} \;\leq\; \hat{v}_j \;\leq\; V_{\text{max},j}^{\circ}. \tag{22}$$

The simplified form in Equation (22) retains only the thermodynamic factor and the capacity scale, and it is the reversible or irreversible $\pm V_{\text{max}}$ box used throughout genome-scale flux balance analysis. It is the tractable baseline representation of a bound when nothing beyond a capacity estimate is known. The factorized form in Equation (7) records which assumptions that baseline makes and which factor requires additional information to relax each of them.

Each class of information just described has a named, established predecessor in the constraint-based modeling literature, and this chapter's contribution is the unifying notation and workflow that connects them, not the invention of any individual constraint. The thermodynamic factor δ_j is the central device of thermodynamics-based flux analysis, which propagates standard free energies and metabolite activity ranges through the whole network to rule out infeasible flux and concentration profiles together [16]. The kinetic factors, $V_{\text{max},j}^{\circ}$ built from a turnover number and an enzyme abundance together with the saturation factor f_j, are the organizing idea behind enzyme-constrained flux balance analysis, most fully realized in the GECKO framework, which folds a proteome-wide capacity constraint into a genome-scale reconstruction [17]. The expression factor (e/e°), transcript and protein balances entered as a modeled quantity rather than a fixed parameter, is the metabolism-and-expression (ME) model construction, which couples the transcription and translation machinery directly to the metabolic network at genome scale [18]. The expression control functions u_j and w_j connect to two established lines of regulatory constraint-based modeling:

regulatory flux balance analysis, which switches expression-linked reactions on and off with Boolean rules derived from a transcriptional regulatory network [19], and PROM, which replaces the Boolean rule with a continuous, data-derived probability of an interaction being active [20]. The allosteric factor θ_j represents the distinct, direct regulation of enzyme activity after the enzyme has been expressed. What the integrative factorization of Equation (7) adds is not a fifth constraint class but a common accounting of how these four already-established devices contribute to one bound, so that a single notation and a single worked derivation carry across all four rather than requiring a separate formalism, and a separate paper, for each.

The expression model above lumped transcription and translation into aggregate capacities $r_{X,j}$ and $r_{L,j}$, but these processes can themselves be resolved to the level of the sequences they read. Transcription of a gene and translation of its protein consume nucleotides, amino acids, and energy in stoichiometric ratios fixed by the gene and protein sequences, and those precursor demands can be computed directly from the sequences and entered into $\mathbf{S}$ as reactions in their own right [5], with the precursors supplied by exchange reactions across the system boundary. Expression then becomes flux-bearing chemistry inside the same linear program as metabolism, a construction carried out for cell-free protein synthesis by Vilkhovoy and coworkers [21], and it is the mechanism behind the cell-free capstone taken up later in this chapter.

5 Worked Example: Urea-Cycle Metabolism

The general bound of Equation (7) integrates four classes of information, and a genome-scale reconstruction may in principle incorporate all four at once. The urea cycle in HL-60 cells, parameterized here with cross-organism public data rather than HL-60-specific measurements, is small enough to make the opposite case concretely: thermodynamic and kinetic information drawn entirely from public databases constrain the prediction without HL-60-specific measurements. The network retains the same steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, Equation (5), and the same linear program, Equation (6), established earlier. What changes from one application to the next is only the content poured into $\mathbf{S}$, $\boldsymbol{\ell}$, $\mathbf{u}$, and $\mathbf{c}$.

The reconstruction couples the four canonical urea-cycle enzymes to a fifth, competing branch and to the extracellular environment (left panel, Fig. 3). Argininosuccinate synthetase (v_1, EC 6.3.4.5) condenses citrulline with aspartate and ATP to argininosuccinate, releasing AMP and diphosphate; argininosuccinate lyase (v_2, EC 4.3.2.1) cleaves that intermediate to arginine and fumarate; arginase I (v_3, EC 3.5.3.1) hydrolyzes arginine to ornithine and urea; and ornithine transcarbamylase (v_4, EC 2.1.3.3) recombines ornithine with carbamoyl phosphate to regenerate citrulline, closing the cycle. A fifth reaction, nitric oxide synthase (v_5, EC 1.14.13.39), draws on the same arginine pool consumed by v_3 and oxidizes it with NADPH and oxygen to nitric oxide and citrulline, diverting flux away from urea production toward signaling chemistry. These five enzymatic reactions act on eighteen metabolites. Fourteen of the eighteen, carbamoyl phosphate, aspartate, ATP, water, oxygen, NADPH, and protons on the uptake side, urea, fumarate, AMP, diphosphate, orthophosphate, NADP$^+$, and nitric oxide on the secretion side, cross the system boundary through fourteen exchange reactions, b_1 through b_{14}, written in the standard secretion-positive convention $M_i \rightarrow \emptyset$, a positive flux denoting secretion and a negative flux uptake. The remaining four, arginine, citrulline, ornithine, and argininosuccinate, are pure cycle intermediates with no exchange reaction of their own: each is produced and consumed entirely inside the network. Collecting five enzymatic and fourteen exchange fluxes against eighteen metabolite balances gives $\mathbf{S} \in \mathbb{R}^{18\times 19}$, $m = 18$ intracellular balances and $n = 19$ reactions.

The thermodynamic factor of Equation (8) was specified first. Standard Gibbs free energies of reaction for all five enzymatic steps were evaluated by eQuilibrator [11]: $\Delta G^\circ_{v_1} = -4.3$, $\Delta G^\circ_{v_2} = -5.5$, $\Delta G^\circ_{v_3} = -51.0$, $\Delta G^\circ_{v_4} = -30.3$, and $\Delta G^\circ_{v_5} = -1220$ kJ/mol. The threshold ΔG^*_j of Equation (8) was fixed here at -10 kJ/mol, a heuristic cutoff rather than one derived from a shown physiological concentration range: a reaction whose standard free energy falls below that cutoff is declared irreversible and its reverse flux forbidden ($\delta_j = 0$), while a reaction whose free energy sits above it is left reversible ($\delta_j = 1$). Only v_1 and v_2 cleared the threshold and remained reversible; v_3, v_4, and v_5 fell below it, v_5 the furthest at -1220 kJ/mol, and were pinned irreversible. Every exchange reaction was left unconstrained by this test and kept a wide default bound in both directions, since the direction of nutrient uptake or product secretion is a matter of medium composition rather than of standard free energy. The kinetic capacity of Equation (9) was specified next. Turnover numbers for the five enzymes were taken from BRENDA [12] and combined with a common reference enzyme abundance $e^\circ = 0.01$ mmol/gDW, a characteristic order of magnitude for intracellular enzyme concentration drawn from compilations such as BioNumbers [13]. Argininosuccinate synthetase and nitric oxide synthase share $k^\circ_{\text{cat}} = 10.0\ \text{s}^{-1}$, giving, after converting the per-second turnover number to the per-hour basis the rest of the chapter uses, $V^\circ_{\max,v_1} = V^\circ_{\max,v_5} = 360\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; arginase I has $k^\circ_{\text{cat}} = 190.0\ \text{s}^{-1}$, giving $V^\circ_{\max,v_3} = 6840\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; and ornithine transcarbamylase has $k^\circ_{\text{cat}} = 410.0\ \text{s}^{-1}$, giving $V^\circ_{\max,v_4} = 14760\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Argininosuccinate lyase has the lowest turnover number: $k^\circ_{\text{cat},v_2} = 3.28\ \text{s}^{-1}$, an order of magnitude below the other four turnover numbers, gives $V^\circ_{\max,v_2} = 118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the smallest of the five capacities by more than threefold. Every exchange reaction was again left with a wide capacity bound, since a boundary reaction represents bulk transport rather than a single characterized enzyme.

With thermodynamic and kinetic information incorporated and the other factors left at their defaults, the general bound of Equation (7) collapses to the two-parameter box of Equation (22) for every enzymatic reaction: no substrate-concentration data enters the model, so the saturation factor is set to $f_j = 1$; enzyme abundance is fixed at the reference value itself, so $(e/e^\circ) = 1$; and no regulatory information is imposed, so $\theta_j = 1$. The bound used in the linear program is therefore $-\delta_j V^\circ_{\max,j} \leq \hat{v}_j \leq V^\circ_{\max,j}$ for v_1 through v_5, with δ_j and $V^\circ_{\max,j}$ as assigned above, and $-1000 \leq \hat{v}_j \leq 1000\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$ for the fourteen exchange reactions, wide enough to be effectively unconstrained. With the feasible box fixed, the one remaining ingredient is the objective, here the maximization of urea export. Exchange reaction b_4 represents urea crossing the system boundary and, following the same secretion-positive convention, is written urea $\rightarrow \emptyset$, so that positive $\hat{v}_{b_4}$ denotes secretion and negative $\hat{v}_{b_4}$ denotes uptake. Maximizing urea export is therefore maximizing $\hat{v}_{b_4}$ directly, so the objective coefficient is set to $c_{b_4} = 1$ and every other entry of $\mathbf{c}$ to zero: maximizing $\mathbf{c}^\top \hat{\mathbf{v}} = \hat{v}_{b_4}$ drives urea secretion as high as the constraints allow. Solving Equation (6) by linear programming [6] with these bounds and this objective gave a unique optimum. The entire enzymatic backbone, v_1 through v_4, saturated at $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the capacity imposed by argininosuccinate lyase, and the nitric oxide synthase branch v_5 carried zero flux: with arginine supply capped by the same v_2 bottleneck that limits the rest of the cycle, none of it was left over to spend on the competing NOS reaction once urea export is the sole objective. The optimal urea export was $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, attained at $\hat{v}_{b_4} = 118.08$, and every exchange flux carrying carbon or nitrogen into or out of the cycle, carbamoyl phosphate and aspartate on the uptake side, urea and fumarate on the secretion side, scaled to match it, giving the full optimal flux distribution (right panel, Fig. 3). An independent flux-variability analysis confirmed this optimum is unique rather than one vertex among several tied solutions: minimizing and then maximizing each of the nineteen fluxes in turn, with the objective held at its optimal value, gave an identical minimum and maximum for every reaction to numerical tolerance, establishing uniqueness for this model and objective.

The single answer above treats every bound as exact, but each was assembled from database estimates that carry their own uncertainty, and propagating that uncertainty turns the point prediction into a distribution. Assigning each turnover number and the reference abundance a lognormal factor of roughly two, each standard free energy a normal spread of a few kilojoules per mole, and four of the five saturation factors (the fifth, f_2 at argininosuccinate lyase, held fixed at one for lack of a measured substrate) a substrate concentration and Michaelis constant drawn from the compilation of Park and coworkers [22], and re-solving the linear program over ten thousand parameter draws by the procedure of Algorithm 1, a Monte Carlo propagation of assumed parameter distributions rather than a bootstrap resampled from replicate measurements, spreads urea export over a right-skewed band with median 105 and mean 156 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and a central ninety-five percent interval of 17 to 614. A capacity-only bootstrap, holding every saturation factor at one and letting only the turnover numbers, the reference abundance, and the standard free energies vary, gives a median urea export of 113 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$, and restoring the four measured saturation factors leaves the median at 105, indicating limited sensitivity to those saturation factors. The width traces almost entirely to the capacity of argininosuccinate lyase: the reversibility switches never bind, because the cycle runs forward at the optimum, and the four saturation factors that Park and coworkers do constrain, for argininosuccinate synthetase, arginase, ornithine transcarbamylase, and nitric oxide synthase, each leave their reaction with capacity to spare above the v_2 bottleneck, so the sampled bound on v_2 dominates the uncertainty in most draws, and the nitric oxide synthase branch, though it carries a small positive mean and a nonzero spread in the rare draws where it turns on, remains negligible overall.

The prediction has limited sensitivity to the measured saturation factors. Specifying f_j with every substrate concentration the database supplies leaves the urea forecast where the capacity scale put it, because the one reaction that sets the answer, argininosuccinate lyase, has no measured substrate: argininosuccinate is a committed cycle intermediate absent from the central-metabolism compilation, so its saturation factor remains at unity while the four measured factors have limited effect. Imputing that missing factor from a physiological argininosuccinate concentration and a BRENDA Michaelis constant, and sampling it alongside the rest, shifts the urea median down to 51 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and widens the band toward zero (Fig. 4), the shift and the spread both governed by the single quantity not measured in the model. Reducing predictive uncertainty would therefore require data for the reaction that binds.

No concentration, expression level, or regulatory parameter specific to HL-60 cells was used: every bound came from a public database, eQuilibrator for the sign of δ_j and BRENDA for the scale of $V^{\circ}_{\max,j}$, combined with a textbook reference enzyme abundance. Two of the four information layers in Equation (7) were specified, and two were left at their default of unity. These assignments produce the flux distribution in Figure 3: a single binding bottleneck at v_2 and a competing branch at v_5 that remains negligible in typical draws, both determined by the reversibility and capacity assigned above. Incorporating the allosteric activity factor θ_j, so that a reaction's capacity depends on the cell's own control state rather than sitting fixed at its database-derived ceiling, is the subject of the feedback-inhibition example that follows.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle example incorporated the thermodynamic and kinetic factors of Equation (7) and held both the expression ratio (e/e°) and the allosteric activity factor θ_j at unity. A first pass at feedback exercises θ_j alone: the fast allosteric inhibition in which the accumulating end product binds the committed enzyme and reduces the catalytic activity of the molecules already present,

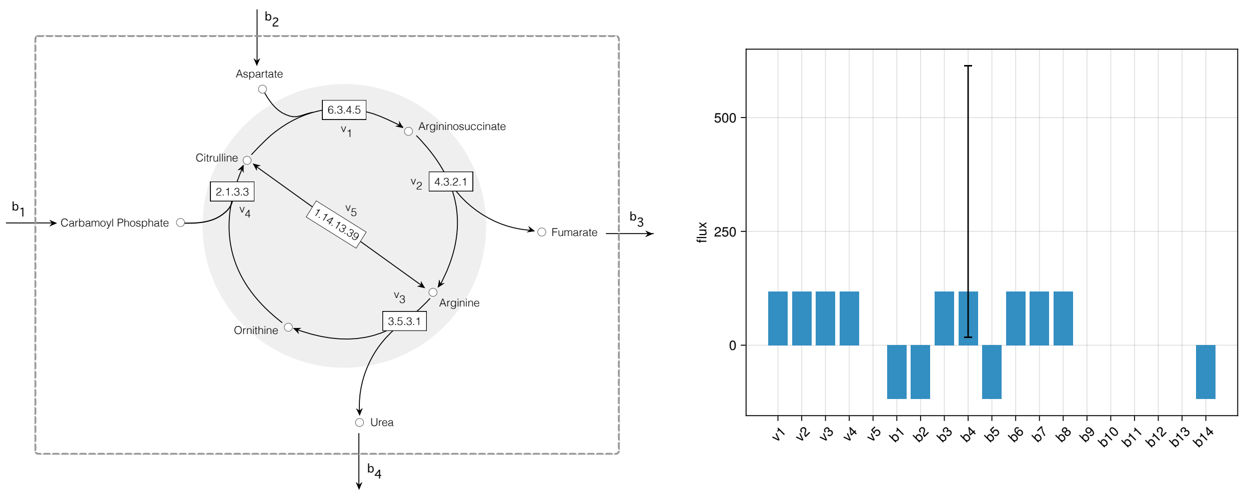

Figure 3: The urea-cycle worked example. Left: the reaction network, with the four cyclic enzymes v_1 through v_4 and the competing nitric oxide synthase branch v_5 labeled by EC number, and the boundary exchanges (carbamoyl phosphate b_1 and aspartate b_2 entering, fumarate b_3 and urea b_4 leaving) crossing the dashed system boundary; arginine, citrulline, ornithine, and argininosuccinate remain internal to the cycle. Right: the optimal flux distribution for the linear program of Equation (6), with bounds set by thermodynamic and kinetic information and the objective $c_{b_4} = 1$ maximizing urea export; bar heights give the flux $\hat{v}_j$ ($\mathrm{mmol\,gDW^{-1}\,h^{-1}}$) at the optimum. Because the cycle is linear, every nonzero backbone flux nearly equals the single throughput up to sign (v_1/v_2 and v_3/v_4 differ from each other only in the rare draws where the nitric oxide synthase branch activates), so the parametric-bootstrap uncertainty of Algorithm 1 is essentially one quantity common to them all; it is drawn as the 2.5–97.5 percentile whisker on the urea-export flux b_4, the reported objective, rather than repeated identically on every bar. The enzymatic backbone v_1 through v_4 saturates at 118.08, the capacity of argininosuccinate lyase (v_2), while the nitric oxide synthase branch v_5 carries zero flux. Exchange fluxes b_1 through b_{14} follow the secretion-positive convention, so positive bars denote net secretion and negative bars net uptake; b_4 (urea) shows the maximized export flux of 118.08 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$.

Algorithm 1 Parametric bootstrap for urea-cycle flux uncertainty

Require: nominal k°_{cat}, e°, ΔG°; nominal saturation inputs $\overline{[S_j]}, \overline{K_{M,j}}$ where measured; spreads $\sigma_{\ln}, \sigma_{\Delta G}$; threshold ΔG^{*}; draws N; seed

seed the random generator

for $i = 1$ to N **do**

  $e \leftarrow e^{\circ} \exp(\sigma_{\ln} Z_e)$ ▷ one shared abundance draw, $Z_e \sim \mathcal{N}(0,1)$

  for each enzymatic reaction j **do**

    $k_{\text{cat},j} \leftarrow k^{\circ}_{\text{cat},j} \exp(\sigma_{\ln} Z_{k,j}), \quad \Delta G_j \leftarrow \Delta G^{\circ}_j + \sigma_{\Delta G} Z_{\Delta G,j}$ ▷ independent $Z_{k,j}, Z_{\Delta G,j} \sim \mathcal{N}(0,1)$ per reaction

    $\delta_j \leftarrow \mathbb{1}[\Delta G_j > \Delta G^{*}]$

    if substrate data measured for reaction j **then**

      $[S_j] \leftarrow \overline{[S_j]} \exp(\sigma_{\ln} Z_{S,j}), \quad K_{M,j} \leftarrow \overline{K_{M,j}} \exp(\sigma_{\ln} Z_{K,j})$ ▷ independent per reaction

      $f_j \leftarrow [S_j]/(K_{M,j} + [S_j])$

    else

      $f_j \leftarrow 1$

    end if

    $V_{\max,j} \leftarrow k_{\text{cat},j}\, e; \quad u_j \leftarrow V_{\max,j} f_j, \quad \ell_j \leftarrow -\delta_j V_{\max,j} f_j$

  end for

  $\hat{\mathbf{v}}^{(i)} \leftarrow \arg\max\{\mathbf{c}^{\top}\mathbf{v} : \mathbf{S}\mathbf{v} = \mathbf{0},\ \boldsymbol{\ell} \le \mathbf{v} \le \mathbf{u}\}$

end for

return per-reaction mean, standard deviation, and 2.5/50/97.5 percentiles of $\{\hat{\mathbf{v}}^{(i)}\}_{i=1}^{N}$

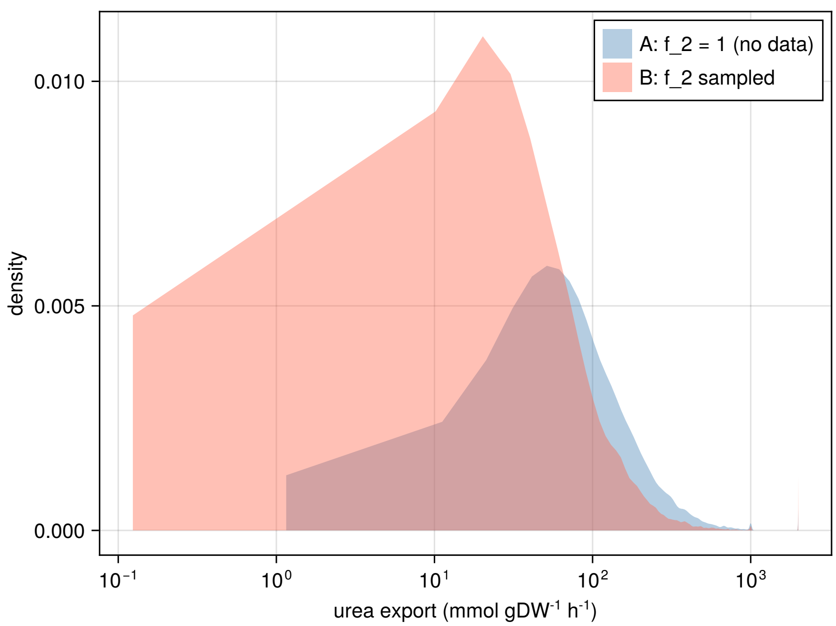

Figure 4: Sensitivity of urea export to saturation information. Configuration A (blue) samples the four saturation factors that Park and coworkers [22] constrain and holds $f_2 = 1$ at the argininosuccinate lyase bottleneck, whose substrate is unmeasured, giving a median urea export of 105 mmol gDW^{-1} h^{-1}. Configuration B (red) imputes f_2 from a physiological argininosuccinate concentration and a BRENDA Michaelis constant and samples it as well, shifting the median to 51 mmol gDW^{-1} h^{-1} and widening the band toward zero. The horizontal axis is logarithmic.

the classical picture of end-product inhibition, with the expression ratio still sitting at unity. A committed biosynthetic step, though, is controlled by its own end product at two levels and on two timescales at once. Alongside the fast allosteric loop the same product represses transcription of the gene encoding the enzyme, a slow loop that lowers the number of enzyme molecules the cell builds rather than the activity of the molecules on hand, and this transcriptional loop is a canonical negative-feedback motif of gene-regulatory circuits, the operon repressed by its own pathway product exactly as the tryptophan operon is repressed by tryptophan. The two loops draw on different currencies, one on catalytic activity and one on enzyme abundance, and they relax on different timescales, so the capacity assigned to the committed step includes both. The pathway kept for the comparison is the following abstract linear chain, which isolates a single control point:

$$\emptyset \xrightarrow{r_0} X_1 \xrightarrow{r_1} X_2 \xrightarrow{r_2} X_3 \xrightarrow{r_3} \emptyset, \tag{23}$$

with r_0 the committed uptake step, r_1 and r_2 interconversions, and r_3 the export step, and the end product X_3 now closing both loops onto r_0 at once: a high steady-state X_3 simultaneously inhibits the activity of the enzyme that catalyzes r_0 and represses transcription of its gene. Incorporating both factors on that single step represents the two feedback mechanisms in the constraint-based model.

The reference behavior was generated from a Biochemical Systems Theory (BST) S-system, the power-law formalism in which every reaction rate is a product of the participating species raised to real-valued kinetic orders [3, 4], but now the committed enzyme is no longer a fixed parameter. Its transcript m and the enzyme protein E_0 are dynamic species carried alongside the three metabolites X_1, X_2, and X_3, five state variables integrated together to a single self-consistent steady state with BSTModelKit.jl [23]. The end product enters the kinetics twice. In the rate of the committed step it appears with a negative kinetic order $-a$, the allosteric activity loop, so that a higher X_3 lowers the throughput the enzyme molecules already present can sustain; in the rate of transcription of the enzyme's gene it appears with a second negative kinetic order $-b$, the expression loop, so that a higher X_3 lowers the rate at which new message, and with it new enzyme, is made. These power laws are local kinetic representations of the two controls. In the notation of Equation (20), the allosteric loop acts through θ_{BST} and the transcriptional loop acts through the control u_j. The partition-function construction developed above provides a bounded active-state representation of the same regulatory roles; the FBA model below uses that representation for its allosteric factor. The two families of balance follow the convention established earlier, the three metabolite balances enforcing $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ with no growth-dilution term while the growth rate μ lives in the transcript and protein balances of Equations (12) and (13), where an intracellular macromolecule is cleared both by its own turnover and by dilution into the growing culture. The committed-step rate carries both controls:

$$v_{r_0} = V^{\circ}_{\mathrm{max},r_0}\,(e/e^{\circ})\,\theta_{\mathrm{BST}}, \qquad \theta_{\mathrm{BST}} = X_3^{-a}, \quad a = 0.4, \qquad \text{transcription} \propto X_3^{-b}, \quad b = 0.6. \tag{24}$$

The expression ratio (e/e°) was not read from a table but computed from the sequences the enzyme is made from. The transcription and translation capacities r_X and r_L follow from a 330-residue enzyme and its 990-nucleotide gene, the polymerase elongating at 60 nucleotides per second and the ribosome at 16 amino acids per second; together with an mRNA half-life of 2.5 minutes, a protein half-life of 35 minutes, and a 60-minute culture doubling time, these fix the residence times of message and enzyme in the steady-state balances of Equations (12) and (13), and with them the reference enzyme abundance e° against which Equation (14) normalizes the ratio (e/e°), the response of that ratio to the end product being carried by the repression order b rather than by the residence times, which cancel from it. One value of μ, the specific growth rate of the culture, enters both balances and is identical in both denominators, $\theta_m + \mu$ for the transcript and $\theta_p + \mu$ for the enzyme; the species-specific first-order turnover θ differs between them. The transcript turns over fast, so that same μ is only about four percent of its total clearance $\theta_m + \mu \approx 0.289\ \text{min}^{-1}$; the enzyme is stable, so the identical μ is more than a third, about thirty-seven percent, of its total clearance $\theta_p + \mu \approx 0.031\ \text{min}^{-1}$. Growth dilution therefore contributes differently to transcript and enzyme clearance even though the same μ appears in both balances.

Because the chain is linear, conservation forces every metabolic reaction to carry one common throughput T, and because the export step r_3 is first order in X_3, that throughput pins the end-product level at $X_3 = T/k_3$. Substituting the two controls back into the committed-step rate, the BST model's allosteric activity factor $\theta_{\text{BST}} = X_3^{-a}$ and the expression ratio the transcriptional repression sets through Equation (14), reduces the coupled five-state system to a single self-consistency condition in X_3 alone, and the value that condition selects is the value the full numerical integration confirmed: $X_3^\star \approx 4$, an allosteric activity factor $\theta^\star_{\text{BST}} \approx 0.574$, an expression ratio $(e/e^\circ)^\star \approx 0.464$, and a throughput $T^\star \approx 2.66$, less than a third of the uninhibited capacity of the committed step (Fig. 5). The control scheme behind those numbers is the standard negative one: a repressor that X_3 corepresses binds the operator of the enzyme's gene and reduces transcription downward as the end product accumulates, but only down to a basal leak floor, the residual expression λ set by the basal occupancy $u^\dagger$ of Equations (12) and (21), so that the gene retains nonzero basal expression. The two feedback strengths were chosen so that each factor is individually load-bearing, $a = 0.4$ on the activity loop and $b = 0.6$ on the expression loop: neither factor alone accounts for the observed reduction, and a model that included one and defaulted the other would produce a different throughput.

The flux-balance model kept the chain's stoichiometry unchanged and placed the entire dual-control content on the single bound of the committed step:

$$0 \ \le\ \hat{v}_{r_0} \ \le\ V^\circ_{\max,r_0}\,(e/e^\circ)\,\theta_{\text{FBA}}(X_3), \qquad V^\circ_{\max,r_0} = 10, \tag{25}$$

the expression and regulatory instance of the general bound of Equation (7), with the thermodynamic switch and the saturation factor left at their defaults and the two remaining factors, expression (e/e°) and activity $\theta_{\text{FBA}}(X_3)$, both in play. The activity factor does not reuse the BST model's kinetic exponent θ_{BST} from Equation (24); a bound built from the exact functional form that generates the quantity it is meant to predict would make the recovery an identity rather than a check. Instead θ_{FBA} is represented by the bounded, two-state Hill occupancy:

$$\theta_{\text{FBA}}(X_3) \ =\ \frac{K^n}{K^n + X_3^n}, \qquad K = 5, \quad n = 2, \tag{26}$$

the repressive two-state specialization of Equation (19): the active configuration has unit weight and the effector-stabilized inactive configuration has weight $(X_3/K)^n$. The parameters K and n were chosen independently of the BST model's allosteric order a rather than fit to reproduce it, and the model assigns no basal active-state weight to the allosteric control, so $\theta_{\text{FBA}} \to 0$ as $X_3 \to \infty$. Both factors were evaluated from the steady-state enzyme abundance and end-product level of the integrated model rather than from its flux. Four solves of the linear program of Equation (6) differed only in this bound. A baseline solve, holding both factors at unity, returned the uninhibited capacity of 10, roughly 3.8 times the BST reference throughput; including the expression factor alone reduced the committed-step capacity to 4.64; including the activity factor alone reduced it to 6.10; and including both returned 2.83, approximating the BST reference value of 2.66 to within about six percent (Fig. 5). This sequence quantifies the contribution of each factor. The stoichiometric matrix is identical whether X_3 closes a single loop, both loops, or neither. Dual control instead enters through the bound on r_0, where the two factors are independent multiplicative factors and each removes a distinct, quantifiable slice of the capacity. The expression and activity factors reduce the uninhibited capacity of 10 to 4.64 and 6.10, respectively, and their product approaches the throughput of the BST reference model.

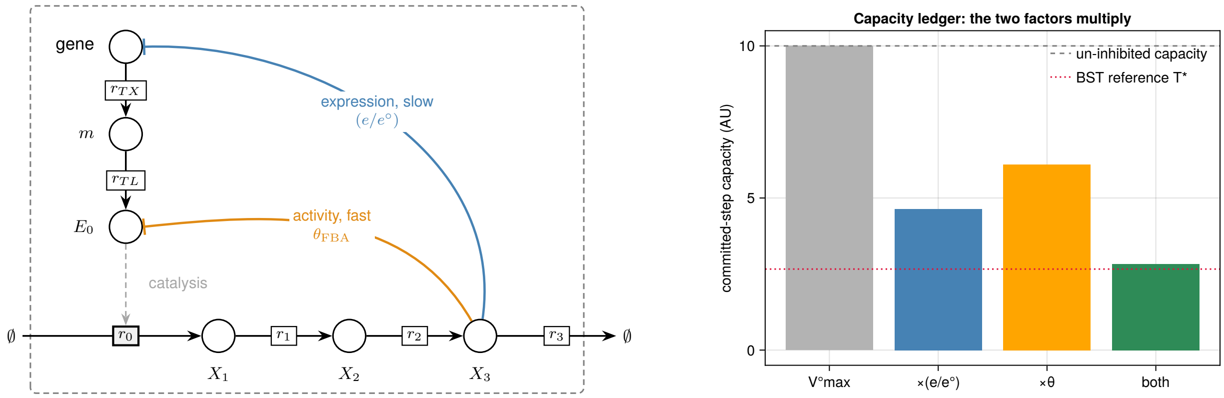

Figure 5: The dual-control feedback worked example. Left: the wiring of the linear chain of Equation (23) inside the dashed cell boundary, with the metabolic backbone r_0 through r_3 along the bottom and the transcription and translation cascade that builds the committed enzyme E_0 down the left. The end product X_3 closes two repressing loops, drawn as blunt-ended arrows, that set the capacity of the committed step r_0: the slow expression loop onto the enzyme's gene, acting through (e/e°) and drawn in blue, and the fast allosteric loop onto the enzyme E_0 already present, acting through θ_{FBA} and drawn in orange, the two colors matching the corresponding capacity reductions at right. Right: the capacity ledger for the committed-step bound of Equation (25). The uninhibited capacity $V^\circ_{\max,r_0} = 10$ is reduced by the expression factor alone to 4.64 (blue) and by the activity factor alone to 6.10 (orange), and only the two factors together approach the throughput of the Biochemical Systems Theory reference, $T^\star \approx 2.66$, landing at 2.83; the dashed line marks the uninhibited capacity.

A single surrogate shape gives one point estimate. The activity factor was assigned the two-state Hill occupancy of Equation (26), a functional form that bears no relation to the power law X_3^{-a} used by the BST reference model, and its half-saturation $K = 5$ and cooperativity $n = 2$ were specified rather than fit. The relevant comparison is therefore whether the reconstruction brackets the BST reference value as the surrogate shape is varied across a plausible window. Sweeping the surrogate over $K \in [3, 8]$ and $n \in [1, 3]$, recomputing θ_{FBA} at the measured X_3, and re-solving the committed-step program at each grid point with both factors included and every other bound held, spread the recovered throughput from about 1.4 to about 4.1 and placed the Biochemical Systems Theory reference value $T^\star \approx 2.66$ inside that band (Fig. 6). The pair $(K, n) = (5, 2)$ that the rest of the construction carries, which returned 2.83, is one interior point of the bracket, and a different shape specification would have produced another value in the same band. No single-factor configuration reached the BST reference value. Across the entire window the activity factor alone held the committed step above about 2.97 and the expression factor alone held it at 4.64 irrespective of K and n, so each single-factor reconstruction sat above the BST reference throughput for every shape while only the two factors together reached down far enough to straddle it. The sweep therefore gives a structural rather than pointwise comparison: including both factors placed the BST reference value inside the reachable range, while including either factor alone did not.

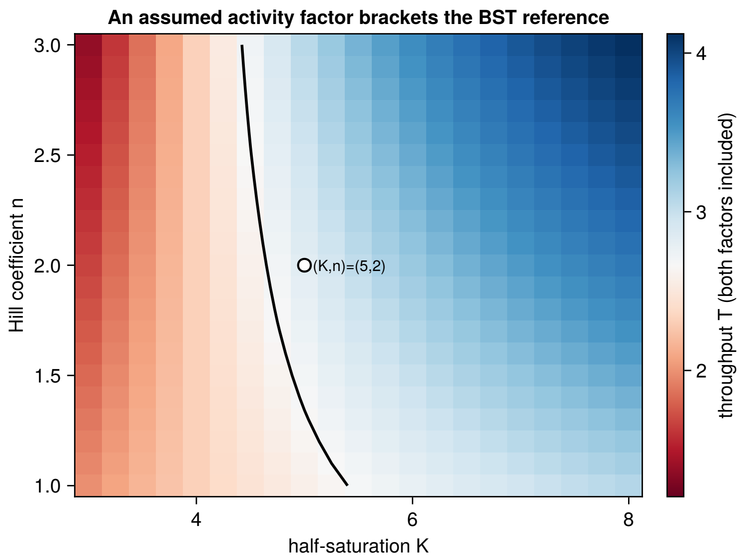

Figure 6: Robustness of the dual-factor recovery to the assumed activity-factor shape. The committed-step throughput with both factors included, $\hat{v}_{r_0} = V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\mathrm{FBA}}$, is evaluated while the assumed Hill shape $\theta_{\mathrm{FBA}} = K^n/(K^n + X_3^n)$ of Equation (26) is swept over its half-saturation K and cooperativity n, with the measured X_3, the expression ratio, and every other bound held. The color scale diverges about the Biochemical Systems Theory reference value $T^{\star} \approx 2.66$, red below and blue above, and the black contour is the locus where the reconstruction equals that value; the recovered throughput runs from about 1.4 to about 4.1, so the reference lies inside the reachable band. The marked point is the shape the rest of the construction carries, $(K, n) = (5, 2)$, returning 2.83. Neither single factor reaches the reference value for any shape in the window: included alone the expression factor holds the step at 4.64 and the activity factor stays above about 2.97, so only the two together bracket $T^{\star}$.

7 Integration in Practice

The urea-cycle and feedback examples incorporated the information layers of Equation (7) one at a time on networks with a single degree of freedom, $n - m = 1$. Two reported systems show those layers operating together at genome scale, where the large right null space makes informative bounds essential (Section 3). The first uses measurements to couple metabolism, expression, and regulation in a dynamic cell-free model; the second modifies bounds to design a glycan-producing strain.

Vilkhovoy and coworkers constructed a sequence-specific model of *E. coli* cell-free protein synthesis in which transcription and translation reactions derived from the target DNA and mRNA sequences enter the stoichiometric matrix alongside metabolism [21]. Building on the promoter-state construction used by Moon, Voigt, and coworkers [14], the effective biophysical model of Adhikari and coworkers supplies the control functions u_j and w_j for promoter and ribosome capacity [15]. This extends the active-subset construction of Equations (15) and (16) from one feedback loop to every regulated step. The resulting dynamic model recomputes thermodynamic, kinetic, expression, and regulatory bounds as the reaction proceeds [8] and is evaluated against trajectories of 63 metabolites together with mRNA, protein, and enzyme-activity data. These measurements also resolve terms omitted from ordinary flux balance analysis. Although cell-free reactions have $\mu = 0$, their molecular pools vary throughout the batch, so the pseudo-steady-state reduction from Equation (1) to Equation (4) does not apply. With no membrane separating the reaction volume from an inferred intracellular compartment, accumulation is observed directly rather than reconstructed from uptake or secretion rates. Cell-free integration therefore provides a setting in which the accumulation term omitted by Equation (5) can be measured directly.

Wayman and coworkers used the framework in the opposite direction. They introduced the N-linked glycosylation pathway of *Campylobacter jejuni* onto a genome-scale reconstruction of *E. coli* metabolism and searched for gene deletions that couple designer-glycan production to growth [9]. Each candidate knockout collapses a reaction bound to zero and defines a new box $[\boldsymbol{\ell}, \mathbf{u}]$ in which the linear program is re-solved before a strain is built. More generally, attenuating or over-expressing an enzyme can lower or raise the (e/e°) factor of Equation (7) without eliminating the reaction. The best single knockout identified by the model increased glycan production roughly threefold when built and tested, although the model selected the deletion rather than predicting the observed yield increase. Here the bounds serve not as a report of cellular behavior but as levers for changing it.

The two capstones thus apply Equation (7) in opposite directions. Measurements refine the cell-free bounds to estimate flux, whereas engineering interventions modify the glycan-model bounds to design it; once the bounds are fixed, both re-solve the same linear program. Design additionally requires choosing which bounds to move, a discrete search over deletions or expression changes that can be formulated through bilevel programming [10]. Estimation and design share the evaluation of a bounded flux model, but only design must first search for the intervention.

8 Outlook

The worked examples populated the factors of Equation (7) from tabulated sources: standard free energies from group-contribution estimates, turnover numbers from enzyme databases, and allosteric and expression controls built from regulatory microstates. The construction does not require those sources. With time-resolved measurements, turnover numbers, saturation constants, allosteric factors θ_j, and expression control functions u_j and w_j can instead be learned from the data they are meant to explain, while tabulated values remain useful as priors when dynamic measurements are unavailable. This approach also extends reaction by reaction to genome-scale reconstructions, where the growing null space of $\mathbf{S}$ makes informative bounds most important: a factor left at unity admits a large subspace of indistinguishable flux vectors and can therefore weaken predictions precisely where stoichiometry constrains them least. The dynamic, data-rich regime of the cell-free capstone is also the natural home for the growth-dilution and accumulation term retained in Equation (4) and omitted in Equation (5). Cell-free reactions are one setting dense enough in time-resolved measurement to estimate fluxes while retaining that term, and the same opportunity extends wherever intracellular state can be tracked closely enough in time, including inside a growing culture. The natural direction is therefore to estimate flux directly against Equation (4), with $\mu\mathbf{x}$ retained wherever measurements support it.

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